

Python Logical Problems – I/O, Data Types, Conditionals, Loops

1. ATM Transaction Validator

Problem Statement

An ATM processes N withdrawal requests sequentially.
Each request has an amount. Rules:

- Withdrawal amount must be a **multiple of 100**
- Account balance must never go negative
- For each transaction, print SUCCESS or FAILED

Input

InitialBalance
N
amount1
amount2
...
amountN

Output

SUCCESS
FAILED
SUCCESS
...
FinalBalance

Sample Input:

5000

4
1200
155
2000
2500

Sample Output:

SUCCESS
FAILED
SUCCESS
FAILED
1800

Hint:

Update balance **only if** both conditions are satisfied.

2. Smart Electricity Billing

Problem Statement

Electricity bill is calculated slab-wise:

- First 100 units → ₹3/unit
- Next 100 units → ₹5/unit
- Remaining → ₹8/unit

If usage > 300 units, add **10% surcharge**.

Input:

units

Output:

total_bill

Sample Input

350:

Sample Output:

1540

Hint:

Apply slabs **incrementally**, then apply surcharge conditionally.

3. Password Strength Evaluator

Problem Statement

Given a password string:

- Must contain at least **1 digit**
- Must contain at least **1 uppercase**
- Length ≥ 8

Print STRONG or WEAK.

Input:

password

Output:

STRONG

Sample Input:

Pass1234

Sample Output:

STRONG

Hint:

Loop through characters and count conditions manually.

4. Traffic Signal Simulation

Problem Statement

A signal cycles every second:

- 1–30 → RED
- 31–45 → YELLOW
- 46–90 → GREEN

Given a time T , print the signal color.

Input:

T

Output:

RED / YELLOW / GREEN

Sample Input:

44

Sample Output:

YELLOW

Hint:

Use modulo arithmetic and range checks.

5. Salary Deduction System

Problem Statement

Employee salary rules:

- Basic salary given
- If late days > 5 → deduct 5%
- If late days > 10 → deduct 10%
- If absent days > 2 → deduct additional 5%

Input:

salary
late_days
absent_days

Output:

final_salary

Sample Input:

50000
8
1

Sample Output:

47500

Hint:

Apply deductions **cumulatively**, not exclusively.

6. Prime Range Analyzer

Problem Statement

Print count of prime numbers between A and B (inclusive).

Input:

A

B

Output:

prime_count

Sample Input:

10

30

Sample Output:

6

Hint:

Check divisibility up to $\sqrt{n}$.

7. Online Order Discount Engine

Problem Statement

Total order amount:

- $\geq 5000 \rightarrow 20\%$ discount
- $\geq 3000 \rightarrow 10\%$
- $\geq 1000 \rightarrow 5\%$
- Else $\rightarrow$ No discount

Print final payable amount.

Input:

amount

Output:

payable_amount

Sample Input:

3200

Sample Output:

2880

Hint:

Apply **only one** highest applicable discount.

8. Binary to Decimal Converter (Without Built-in)

Problem Statement

Given a binary number, convert to decimal.

Input:

binary_number

Output:

decimal_number

Sample Input:

101101

Sample Output:

45

Hint:

Process digits from right to left using powers of 2.

9. Mobile Battery Drain Simulator

Problem Statement

Battery starts at 100%. Each app drains fixed % per minute.
Stop when battery ≤ 0 . Print minutes used.

Input:

drain_per_minute

Output:

minutes

Sample Input

7

Sample Output

15

Hint:

Use a loop until battery ≤ 0 .

10. Exam Result Processor

Problem Statement

Input marks for 5 subjects:

- If any mark $< 35 \rightarrow$ FAIL
- Else average $\geq 75 \rightarrow$ DISTINCTION
- Else PASS

Input

m1 m2 m3 m4 m5

Output

FAIL / PASS / DISTINCTION

Sample Input

80 78 74 90 88

Sample Output

DISTINCTION

Hint:

First validate failure condition, then classify.

11. Number Compression Counter

Problem Statement

Given a number, count how many times it can be divided by 2 until it becomes odd.

Input

number

Output

count

Sample Input

40

Sample Output

3

Hint:

Use a loop and modulo check.

12. Vowel Frequency Analyzer

Problem Statement

Count total vowels in a given sentence (case-insensitive).

Input

sentence

Output

vowel_count

Sample Input

I Love Python

Sample Output

4

Hint:

Normalize case before comparison.

13. Train Ticket Fare Calculator

Problem Statement

Fare rules:

- Distance $\times$ ₹2/km
- Senior citizen $\rightarrow$ 30% discount
- Child (<12) $\rightarrow$ 50% discount

Input

distance

age

Output

fare

Sample Input

200

65

Sample Output

280

Hint:

Calculate base fare first, then apply age-based rule.

14. Number Pattern Validator

Problem Statement

Check if digits of a number are **strictly increasing** left to right.

Input

number

Output

YES / NO

Sample Input

13579

Sample Output

YES

Hint:

Compare adjacent digits.

15. Smart Door Lock System

Problem Statement

User gets 3 attempts to enter correct PIN.

- Correct → ACCESS GRANTED
- All wrong → LOCKED

Input

correct_pin
attempt1
attempt2
attempt3

Output

ACCESS GRANTED / LOCKED

Sample Input

4321
1111
2222
4321

Sample Output

ACCESS GRANTED

Hint:

Exit loop early on success.

16. Water Tank Overflow Detector

Problem Statement

Tank capacity is 1000L. Inflow every minute given.
Stop when overflow occurs and print minute number.

Input

N
inflow1 inflow2 ... inflowN

Output

overflow_minute

Sample Input

5
200 300 250 400 100

Sample Output

4

Hint :

Accumulate volume gradually.

17. Armstrong Number Checker

Problem Statement

Check if a number equals sum of cubes of its digits.

Input

number

Output

YES / NO

Sample Input

153

Sample Output

YES

Hint:

Extract digits using modulo and division.

18. Bus Seat Allocation

Problem Statement

Bus has 40 seats. For each booking request:

- If seats available → CONFIRMED
- Else → WAITLISTED

Input

N
request1
request2
...

Output

CONFIRMED / WAITLISTED

Sample Input

3
15
10
20

Sample Output

CONFIRMED
CONFIRMED
WAITLISTED

Hint:

Track remaining seats.

19. Number Mirror Validator

Problem Statement

Reverse a number and check if original equals reversed.

Input

number

Output

PALINDROME / NOT PALINDROME

Sample Input

1221

Sample Output

PALINDROME

Hint:

Build reverse using arithmetic.

20. Digital Lock Countdown

Problem Statement

Given a number, repeatedly subtract sum of its digits until result is a single digit.

Input

number

Output

final_digit

Sample Input

987

Sample Output

6

Hint:

Nested loops: digit sum inside reduction loop.